

**Layer 1
Q-Matrix
Construction**

2,643 Math Problems
(GSM8K + MATH)

LLM Metacognition
→ 7,723 Skill Labels

SBERT
Embedding

HAC Clustering
(K = 50)

Binary Q-Matrix
(2,643 × 50)

**Layer 2
Skill
Profiling**

Response Matrix
(235 × 2,643)
+ Q-Matrix

NCDM
Training

Skill Mastery Profiles
(235 × 50)

**Layer 3
Routing**

New Question

SBERT
Embedding

Text-Conditioned
NCDM

P(correct)
per LLM

Route to
Best LLM